

JARVIS

Software Requirements Document

Retail Stock Trading Application

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1. Introduction

1.1 Executive Summary

The Retail Stock Trading Application is a digital platform designed to allow retail investors to buy and sell stocks directly using a mobile or web application. Traditionally, many customers rely on brokers to execute stock trades, which requires phone calls, appointments, or interactions with third-party trading platforms. This process often introduces delays and makes trading less accessible to everyday investors.

The goal of this project is to create a simple and accessible trading platform that allows users to independently manage their investments. Through the application, users will be able to search for stocks, monitor market information, place buy or sell orders, and track the performance of their investment portfolios. The system is intended to be user-friendly so that even individuals with limited technical knowledge can understand and use the platform effectively.

The first version of the system will focus on a Minimum Viable Product (MVP) that supports basic trading functionality for stocks listed on North American exchanges. More advanced financial instruments such as cryptocurrency trading, leveraged trading, options trading, or foreign exchange markets will not be included in the initial release but may be considered in future expansions of the platform.

1.2 Purpose

The purpose of this Software Requirements Document is to define the functional and non-functional requirements of the Retail Stock Trading Application. This document serves as a communication tool between stakeholders, business analysts, developers, and system architects. It ensures that everyone involved in the project has a clear understanding of the system's capabilities, constraints, and expectations.

The requirements described in this document were gathered during the elicitation meeting with the project stakeholder. These requirements will guide the design, development, testing, and deployment of the trading platform.

1.3 Intended Audience

This document is intended for multiple stakeholders involved in the project. Business stakeholders will use it to verify that the system aligns with business goals. Software developers will reference it while implementing system features. Quality assurance teams will use it to validate that the system behaves according to the requirements. DevOps teams will rely on this

document when deploying and maintaining the infrastructure. Security and compliance teams may also review the document to ensure that the system meets financial data protection and privacy regulations.

1.4 Scope

The Retail Stock Trading Application will provide users with the ability to trade stocks through a secure digital platform connected to their existing bank accounts. The system will allow users to authenticate using their banking credentials, search for stocks, place trading orders, and monitor their portfolio performance.

Although the application connects to existing bank accounts for funding purposes, it will operate as an independent trading platform rather than a feature embedded directly inside the bank's core system. The application will retrieve stock market information from external financial data providers and will maintain a record of user transactions and portfolio positions.

The initial release will focus on basic trading functionality and portfolio monitoring. Additional trading instruments and advanced financial features may be introduced in future versions of the system.

2. Overall Description

2.1 Business Context

The organization intends to develop a digital trading platform that allows retail customers to participate in stock trading without relying on traditional broker-assisted services. Many customers currently use third-party trading platforms to manage their investments, which results in lost trading activity and reduced customer engagement within the organization's ecosystem.

By offering its own trading platform, the organization can provide additional financial services to its customers while also capturing trading fees and gaining insights into customer investment behavior. The system is designed to reduce friction for users by allowing them to quickly access market information and execute trades independently.

2.2 User Needs

Retail investors require a platform that allows them to manage their investments easily and securely. Users should be able to log in quickly, search for stocks, and execute trades without needing assistance from a broker. In addition, they need the ability to view their portfolio performance and track how their investments change over time.

The application should also be designed with accessibility in mind so that individuals who are not highly technical can still use it comfortably. Clear portfolio information, intuitive navigation, and simple trading workflows will help ensure that the platform is usable for a wide range of investors.

2.3 Assumptions and Dependencies

Several assumptions were identified during the requirements gathering process. The system assumes that users already have an existing bank account that can be used to fund trading activities. Authentication will rely on existing bank credentials rather than storing separate login information within the trading platform.

The system will depend on external APIs to retrieve stock market data, including price information and historical performance. Cloud infrastructure will be used to support scalability and high availability. The platform must also integrate with backend services responsible for processing trade requests and storing transaction records.

3. System Overview

The Retail Stock Trading Application will consist of multiple system components that work together to deliver trading functionality. Users will access the system through either a mobile application or a web interface. Once authenticated, users will be able to search for stocks, view market data, and submit trading orders.

When a user places an order, the request will be transmitted to backend trading services responsible for validating and processing the transaction. These services will record the trade in the system database and update the user's portfolio accordingly. The system will also communicate with external financial data providers to retrieve stock price information used for displaying charts and calculating portfolio valuations.

4. User Stories

A retail investor wants to log in to the application using their bank credentials so that they can securely access their investment account. The system should support additional authentication methods such as multi-factor authentication or biometric verification to enhance security.

A user wants to search for stocks and view price information before making an investment decision. The application should display relevant market information, including historical price charts and trading statistics.

An investor wants to monitor the performance of their portfolio over time. The system should calculate the current value of their investments and display gains or losses relative to the purchase price.

A user wants to receive notifications about significant market changes or trading activity so that they can respond quickly to investment opportunities.

5. Stakeholders

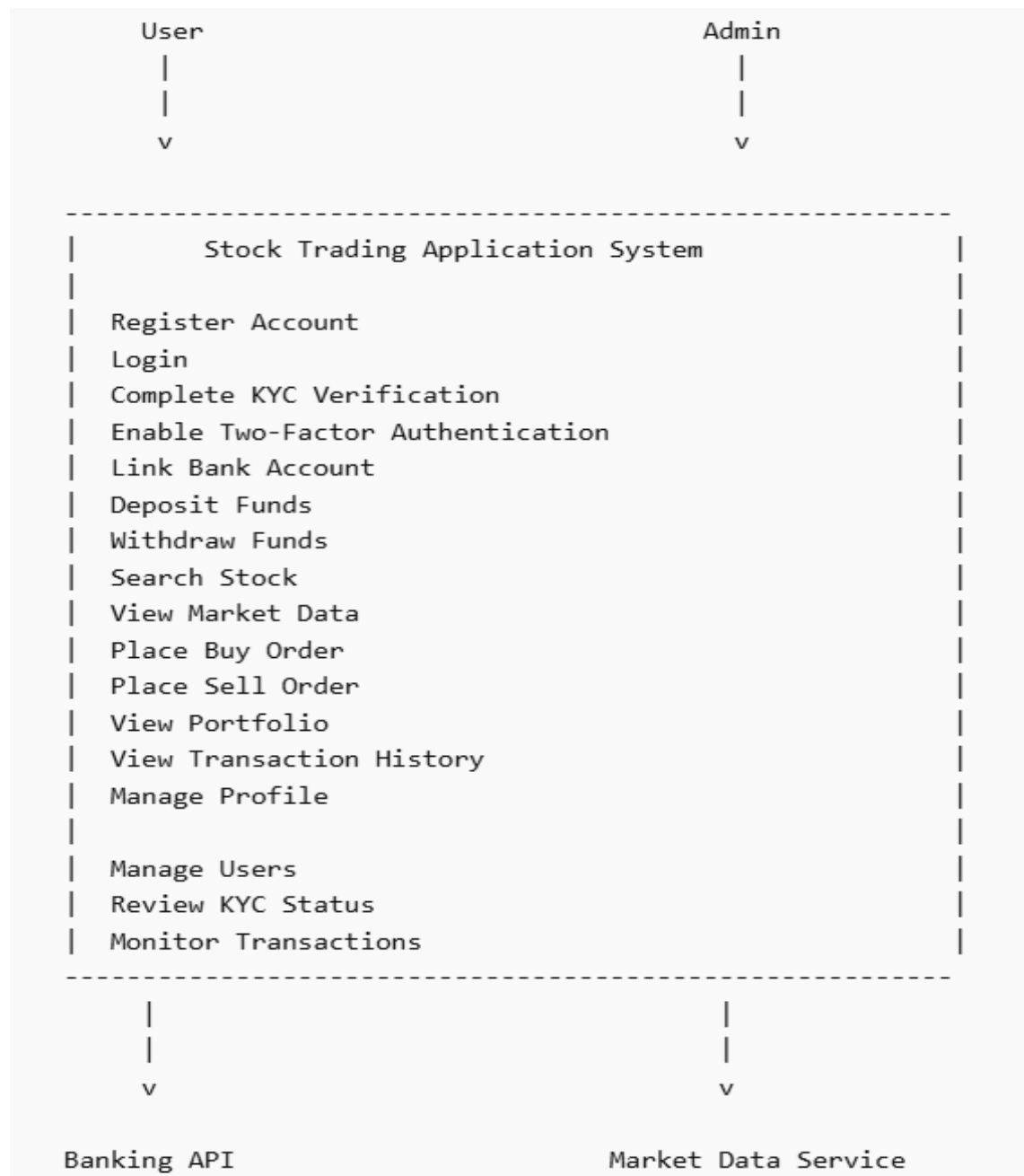
The primary users of the system are retail investors who will use the application to manage their stock investments. Jarvis Consulting Group serves as the organization responsible for developing the platform. Trevor Doto acts as the project stakeholder overseeing the requirements and final delivery of the system.

Additional stakeholders include the software development team responsible for implementing system functionality, the quality assurance team responsible for testing, the DevOps team responsible for deployment and infrastructure, and security and compliance teams responsible for ensuring that financial regulations and privacy standards are satisfied.

6. Use Case Diagram

The Use Case Diagram illustrates how different actors interact with the Stock Trading Application System. The primary actor is the **User**, who performs the main activities such as registering an account, logging in, completing KYC verification, linking a bank account, depositing or withdrawing funds, searching for stocks, viewing market data, placing buy or sell orders, monitoring their portfolio, viewing transaction history, and managing their profile.

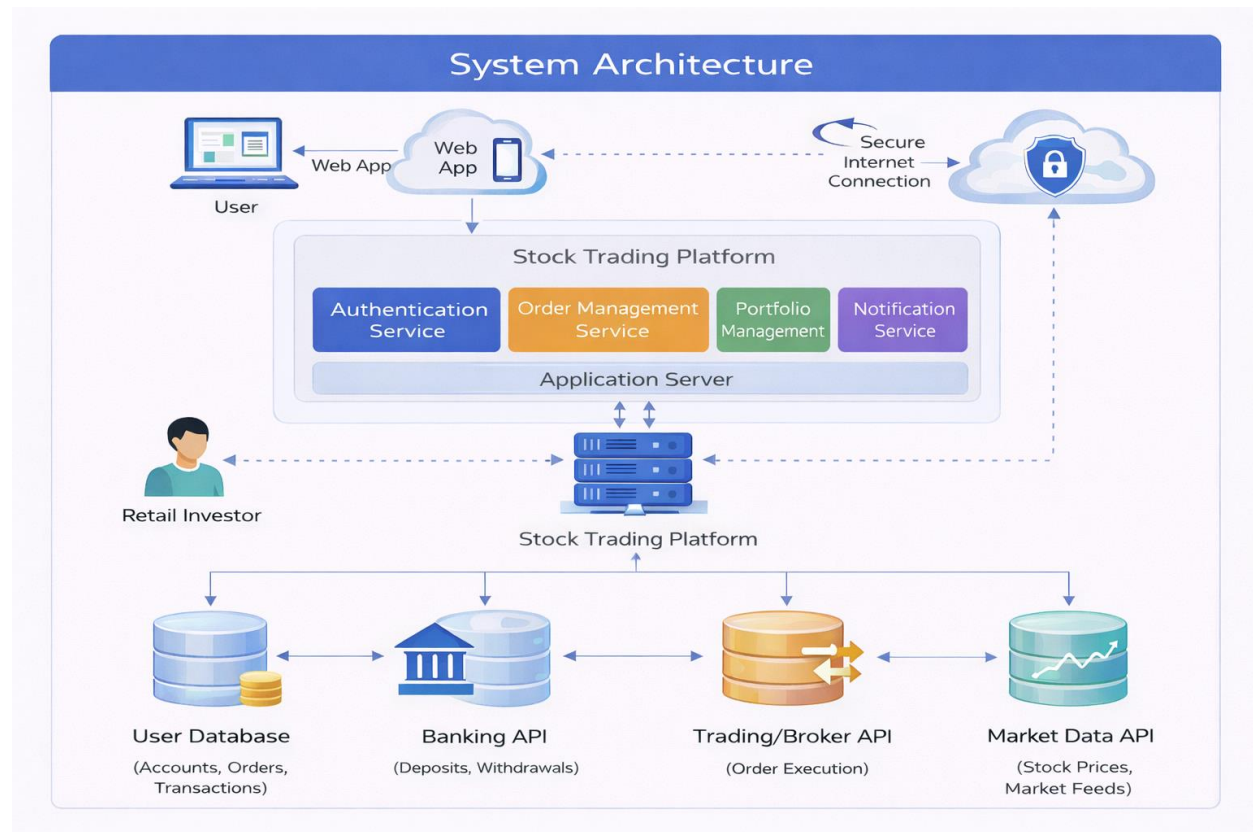
The system also includes an **Admin** actor responsible for administrative tasks such as managing users, reviewing KYC status, and monitoring transactions. In addition, the application interacts with external services. The **Banking API** supports bank account linking and fund transfers, while the **Market Data Service** provides stock prices and market information required for trading.



7. System Architecture

The architecture of the system includes several interconnected components. Users interact with the platform through mobile or web interfaces that communicate with backend services through secure APIs. These backend services handle authentication, trade processing, portfolio calculations, and notification delivery.

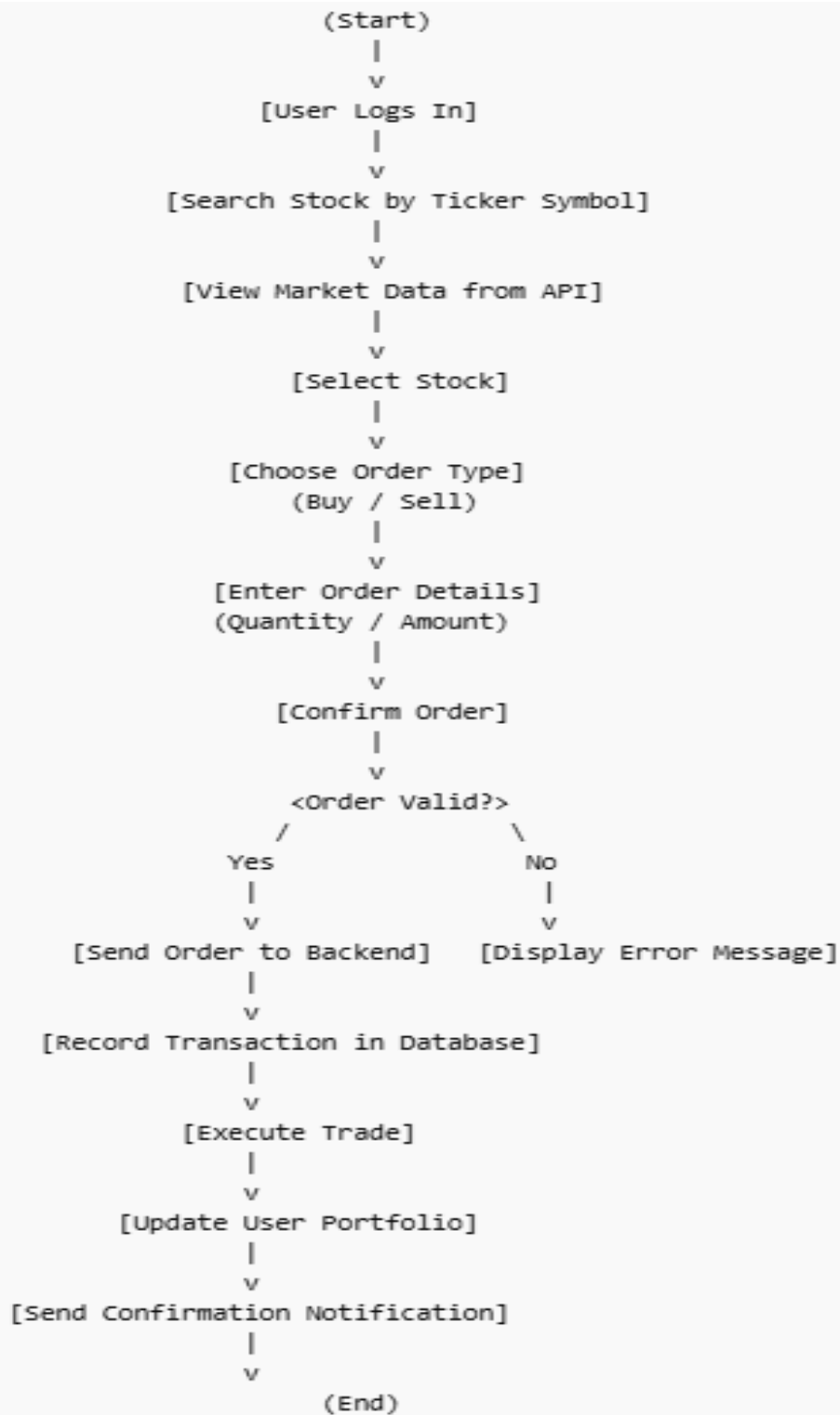
The system will store trading data and user portfolio information in a database while retrieving stock market information from external financial APIs. Additional services may handle monitoring, logging, and infrastructure management to ensure the system remains stable and secure.



8. Trading Workflow

When a user wants to execute a trade, the process begins with the user logging into the trading application. The user searches for a stock using its ticker symbol and views market data retrieved from external financial APIs. After reviewing the stock information, the user selects whether they want to buy or sell shares.

The user specifies the number of shares or investment amount and confirms the trade request. The order is then transmitted to backend trading services, where it is validated and recorded in the system database. Once the transaction is processed, the system updates the user's portfolio and sends a confirmation notification.



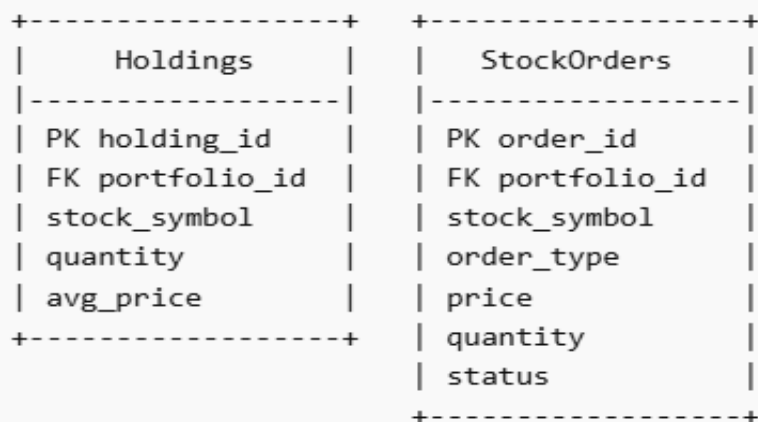
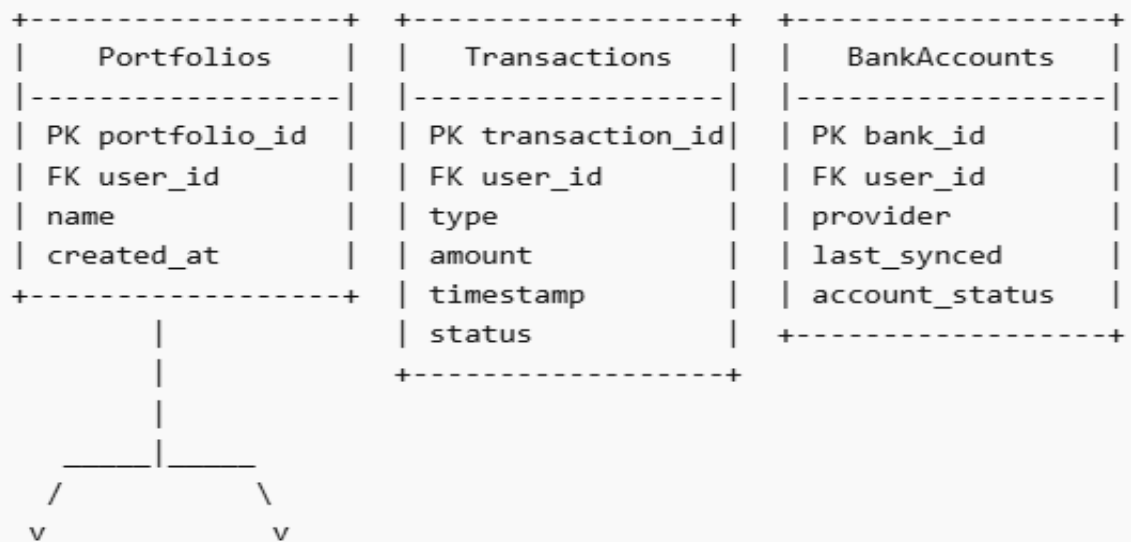
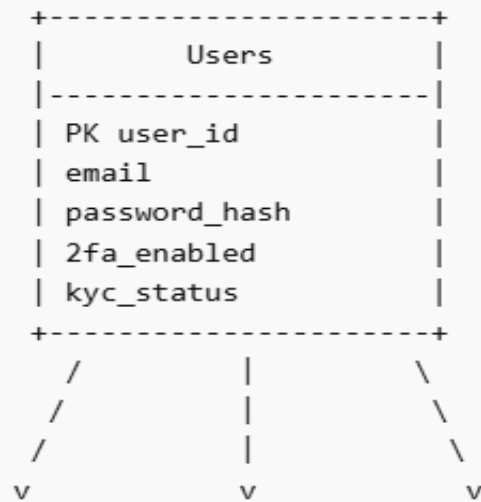
9. Data Model (ER Diagram)

The ER Diagram represents the core data structure used by the Retail Stock Trading Application. The system manages several key entities, including Users, Portfolios, Transactions, BankAccounts, Holdings, and StockOrders. These entities store the information required to manage user accounts, portfolio positions, and trading activity.

The Users entity stores account information such as email, password hash, two-factor authentication status, and KYC verification. Each user may have multiple BankAccounts, Portfolios, and Transactions. Bank accounts represent the funding sources connected to the user.

Each Portfolio belongs to a user and contains multiple Holdings, which represent the quantity and average price of stocks owned. When a user places a trade, the system creates a StockOrders record linked to the portfolio, storing information such as stock symbol, order type, price, quantity, and status. Transactions maintain a record of financial activity within the system.

These relationships allow the platform to track user investments, trading history, and portfolio performance in a structured and reliable way.



10. Functional Requirements

The system must allow users to authenticate using their existing bank credentials and support additional authentication mechanisms such as multi-factor authentication and biometric login. Users must be able to search for stocks using ticker symbols and retrieve price information from external financial data providers.

The system must allow users to place buy or sell orders for supported securities and record all trading transactions in the system database. It must also display the historical trading activity of users and calculate portfolio valuation based on current market prices.

Users should have the ability to monitor stocks of interest through watchlists and receive notifications related to trading activity or significant market movements.

11. Data Requirements

The trading platform must maintain records of user trading activity and portfolio positions. Each transaction should include information such as the stock symbol, trade date, number of shares purchased or sold, and execution price. These records will be used to calculate portfolio performance and generate historical trading reports.

User information will be referenced through identifiers associated with existing bank accounts rather than duplicating sensitive personal data inside the trading application. Market price information will be retrieved from external APIs and may be temporarily cached for performance purposes.

12. Non-Functional Requirements

The system must provide reliable performance and scalability to support a large number of users accessing the platform simultaneously. The application should be able to support up to one million registered users while maintaining fast response times during trading operations.

High system availability is critical because users must be able to access the application during market hours. The platform should maintain uptime close to 99 percent or higher. Security requirements include encryption of sensitive financial data, secure authentication mechanisms, and compliance with privacy regulations such as PIPEDA. The application should also support accessibility features such as screen readers to ensure that the platform can be used by individuals with disabilities.

13. Infrastructure and Monitoring

The application will be deployed on cloud infrastructure, likely using Amazon Web Services or a similar enterprise cloud provider. Monitoring tools will be used to track system performance, detect failures, and ensure high availability of the platform. Logging and monitoring services will help the operations team maintain system reliability and investigate potential issues.

14. Schedule, Timeline, and Milestones

The estimated completion time for the Retail Stock Trading Application project is approximately three weeks for the initial prototype and documentation phase. During the first week, the focus will be on gathering and refining requirements, completing the Software Requirements Document, and defining the system architecture, use case diagram, and data model. The second week will focus on implementing the core system design and validating the trading workflow and functional requirements. The third week may be used for testing, final revisions, and stakeholder review. The primary milestone of this project is the delivery of a working system design and requirements document that demonstrates the core functionality of the trading platform.

15. Future Enhancements

While the initial release focuses on basic stock trading functionality, the system architecture will allow additional features to be introduced in the future. These enhancements may include support for cryptocurrency trading, options trading, leveraged investments, and foreign exchange markets. Designing the system with extensibility in mind will allow the organization to expand the platform without major architectural changes.